Mathieu Bonneau

M.Sc. Computer Science (AI) — AI Developer: LLM applications, RAG and agents

Available from January 2027

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PROFILE

Computer science master's student in AI who bridges software development and AI. I build LLM applications end to end (RAG, LangGraph agents, access control) and take them from prototype to deployed services with FastAPI, Docker and Azure, with tests, cost monitoring and guardrails. I compare frameworks and models on measured criteria and state their limits plainly.

EXPERIENCE

Research Intern — Human Activity Recognition Jul – Dec 2026

LIARA Lab — Université du Québec à Chicoutimi · Chicoutimi, QC

– Worked autonomously on an open research topic: reviewed self-supervised architectures (I-JEPA, V-JEPA), compared approaches and made recommendations to the team.
– Built reproducible ROS and Python pipelines to record, synchronize, calibrate and replay multi-sensor LiDAR streams.

PROJECTS

RAG Chatbot with Role-Based Access Control 2026

RAG · FastAPI · ChromaDB · Docker · Azure | Live demo · GitHub

– Internal chatbot over 5 departments where each of 6 roles only retrieves the chunks it may read, so the LLM never sees restricted content.
– Out-of-scope classifier, PII redaction, token and cost logging with alerts, 53 pytest tests; API on Azure Container Apps (scale-to-zero), Streamlit UI.

RAG Frameworks Comparison 2026

LlamaIndex · LangChain · Haystack · FAISS · RAGAS | Live demo · GitHub

– Same pipeline built three times on one FAISS index; LlamaIndex's default metadata embedding cut the right record's similarity from 0.575 to 0.396.
– 20 hand-reviewed test questions, LLM judge (17/20, 20/20, 20/20 correct) and RAGAS; role filtering per framework (built-in filters or a custom postprocessor).

Voice AI Customer-Support Agent 2026

LangGraph · FastAPI · ElevenLabs · Azure | GitHub

– LangGraph agent routing intents to RAG FAQ, transaction lookup or callback scheduling, behind an OpenAI-compatible endpoint for ElevenLabs voice calls.
– Deployed on Azure Container Apps (scale-to-zero); documented the cost vs. cold-start latency trade-off.

Multi-Agent Coding Assistant 2026

LangGraph · Groq · Pydantic · Streamlit | Live demo · GitHub

– Planner → Architect → Coder pipeline turning a plain-English request into a multi-file project, with Pydantic-validated JSON outputs and retries.

EDUCATION

Université du Québec à Chicoutimi (UQAC) 2025 – present

M.Sc. Computer Science — Artificial Intelligence · Chicoutimi, QC

Télécom Saint-Étienne 2023 – present

Engineering degree — image processing · Saint-Étienne, France

TECHNICAL SKILLS

Programming
Python, SQL, JavaScript, Java, C++

LLM applications
OpenAI-compatible APIs (Groq, Ollama); prompt design, structured outputs, guardrails

RAG
LlamaIndex, LangChain, Haystack; FAISS, ChromaDB; chunking, metadata filtering

Agents
LangGraph; tool calling, intent routing, multi-agent pipelines

Evaluation
RAGAS, LLM-as-judge, retrieval metrics, synthetic test sets

Backend & deployment
FastAPI, REST, pytest, Docker, Azure Container Apps, Streamlit Cloud

Production
Access control, PII redaction, token and cost monitoring, secrets management

Machine learning
PyTorch, scikit-learn, Pandas, NumPy; computer vision (OpenCV)

Tools & methods
Git, GitHub, Linux, uv; Agile/Scrum

LANGUAGES

French
Native

English
Professional working proficiency — TOEIC (B2)

INFORMATION

Willing to relocate to Québec City or Montréal

Driver's licence — France, category B